Panning and Sequencing

Intro to Audio Tech 1
Jocelyn Kavanagh
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Georgia Tech

Lesson Outline

- Quiz 1
- Panning
- Sequencing and Lists in Max
- Up Next…

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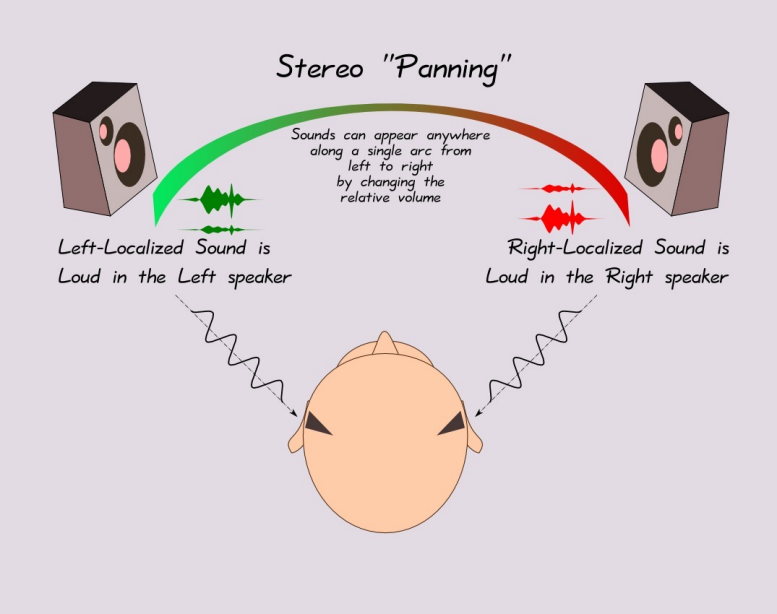
Quiz 1

- You can use a calculator or your phone/laptop calculator
- You do not have to answer the last question in complete sentences, bullet points are fine

Panning

Panning

- Placement of sound within the spatial environment created by two or more channels
- We'll work with stereo (L/R) panning for this class
- We'll model panning by representing position between channels as X, a value between 0-1
  - 0 is 100% of the sound on the left
  - 0.5 is sound in the center or 50% left and 50% right
  - 1 is 100% of the sound on the right



Linear Panning in Practice

- We can multiply amplitude *a* by our panning multiplier *X* so that
  - $a * (1 - X)$ is left presence
  - $a * (X)$ is right presence
- This works well for left and right, but if we are at the center, we send 50% to the left and 50% to the right causing the intensity to be lower
  - ”Hole in the middle” effect

Reminder: Intensity, Power, Amplitude

- Power is the total energy of a signal
- Intensity is the energy transmitted through a given area over time
- Power and Intensity are proportional in a fixed area
- Power is proportional to amplitude squared

$$P \propto I$$

$$P \propto a^2$$

Constant Power Panning

- We can use this relationship to prove why we get the “hole in the middle” effect

$$P_{left} = {a_{left}}^2 = 0.5^2 = 0.25$$
$$P_{right} = {a_{right}}^2 = 0.5^2 = 0.25$$
$$P_{tot} = P_{left} + P_{right} = 0.5$$

$$P_{left} = {a_{left}}^2 = 0^2 = 0$$
$$P_{right} = {a_{right}}^2 = 1^2 = 1$$
$$P_{tot} = P_{left} + P_{right} = 1$$

Constant Power Panning

- We can also use this relationship to create a new model for panning

$$Left = a * \sqrt{1 - X}$$
$$Right = a * \sqrt{X}$$

Now…

$$P_{left} = {a_{left}}^2 = \sqrt{1 - 0.5}^2 = 0.5$$
$$P_{right} = {a_{right}}^2 = \sqrt{0.5}^2 = 0.5$$
$$P_{tot} = P_{left} + P_{right} = 1$$

Lists and Sequencing in Max

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Homework 1 – Phase Music

- Make a patch that plays two copies ("voices") of a series of pitches, looping at slightly different speeds so that they slip in and out of phase
- The two copies should be identical except that one runs through its series of pitches slightly faster than the other

- What are some of the elements we'll need for this patch?
  - A way to keep time between the sequences
  - List(s) of pitch values
  - A way to iterate through the pitch values
  - Oscillators and DAC

Controlling Time

- Metro outputs a bang every x milliseconds when triggered

- Tempo outputs a bang based on a metrical structure and bpm

- Delay delays a bang after x milliseconds

- Pipe delays a number/list/symbol after x milliseconds

- Counter keeps count and outputs the count with every bang

Exploring Lists in Max

- In groups, explore the following Max objects, and how they may be helpful for your homework:
  - Coll
  - zl (specifically nth, mth, splice, lookup)
  - Route
  - itable
- We'll regroup and present these objects to each other

Up Next…

- Readings/Videos!
- HW 1 is due Sept 12 at 11:59p
- Exercise 3 due on Monday